

Bilgin

December 9th, 2019

Total points: 100

Please fill in your answers in the space supplied

Show all of your calculations, conversions and work to receive full credit.

QUESTION 1 (15 pts)

True or False (2 pts each)

_____ The upstream and downstream enthalpies of flow through an insulated valve is equal.

_____ Entropy, S , is path-dependent.

_____ Gibbs energy is an intensive property.

_____ A closed system can experience an increase in entropy only when there is heat transfer from the system.

_____ The change in Gibbs free energy (dg) is equal to zero for a phase change from a saturated liquid to a saturated vapor.

_____ The second Carnot corollary states that reversible power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

Fill in the blank

(3 pts) A gas goes from State A to State B in a reversible, adiabatic process. It then goes from State B back to State A by a different pathway that is irreversible and not adiabatic. The entropy change, (ΔS) for the irreversible pathway is

_____ (greater than/equal to/ less than) zero. Explain.

QUESTION 2 (10 pts)

- a) (5 pts) A steam engine operates with a theoretical maximum efficiency of %50 and a minimum temperature of 150°C. Calculate temperature of the engine to maintain maximum efficiency.

- b) (5 pts) The Reynolds number, Re , is a dimensionless number that measures the ratio of inertial forces to viscous forces in fluid flow;

$$Re = \frac{\rho V L}{\mu}$$

In this equation, ρ is the density (kg/m^3), V is the velocity (m/s), L is a characteristic length (m), and μ is the viscosity. What are the dimensions of μ ?

Show all your work for full credit.

QUESTION 3 (10 pts)

- a) (2 pts) Write the specific Maxwell relation that corresponds to the thermodynamic function:

u (s, v): _____

- b) (2 pts) Write the two, second mixed partial derivatives for the exact differential:

dh = Tds + vdp : _____

- b) (6 pts) A 3 kg mass of saturated liquid water has a pressure of 10.0 bar. What is the Helmholtz energy, in kJ?

Show all your work for full credit

QUESTION 4: (32 pts)

3 kg of water, initially a saturated liquid at 120°C is contained piston-cylinder device. The water undergoes a process. Heat is transferred between the piston-cylinder and a 25°C room until the water becomes a state where $x=0.8$. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored.

- a) (2 pts) Draw a schematic of the system and label the system boundary.
- b) (3 pts) State the engineering model
- c) (4 pts) Write the simplified energy balance and simplified entropy balance.
- d) (8 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.
- e) (5 pts) Determine the change in entropy of the water during the process, **ΔS , in kJ/K**
- f) (5 pts) Determine the entropy generated during the process, **σ , in kJ/K**. State if this process is reversible, irreversible or impossible.
- g) (5 pts) Carefully sketch the process on P-v and T-s diagram.

QUESTION 5: (33 pts)

A cogeneration system operates at steady state. Water vapor enters the system at 20 bar, 360°C, velocity of 50m/s and mass flow rate of 1.5kg/s. Power is developed by the system at a rate of 2.2MW. There are two outlets for the system: process steam and hot water. Process steam leaves as saturated vapor at 1 bar, 100m/s, and hot water leaves as saturated liquid at 1 bar, 100m/s. Mass flow rates of process steam and hot water are equal. Neglect any changes in potential energy.

- a) (4pts) Draw a schematic of the system and label the system boundary.
- b) (5pts) State the engineering model.
- c) (6pts) Write simplified balances for mass, energy and entropy.
- d) (10pts) Calculate the rate of heat transfer, in MW, between the system and its surroundings.
- e) (8pts) Calculate the entropy production for the process, in kJ/K-s.

